III Semester B.Tech
Data Analytics
CSS 2103

Q1. The heights of six randomly chosen sailors are in inches: 63, 65, 58, 69, 71, and 72. The heights of ten randomly chosen soldiers are in inches: 61, 62, 65, 66, 69, 69, 70, 71, 72, and 73. Do these figures indicate that the soldiers are on average, shorter than the sailor? Statistically test at 5% significance level . (4)

Answer:

Step 1: State Hypotheses (0.5 Mark)

- Null hypothesis (H_0): $\mu_{soldiers} = \mu_{sailors}$
- Alternative hypothesis (H_1): $\mu_{soldiers} < \mu_{sailors}$

Step 2: Sample Means (0.5 Mark)

$$\bar{x}_1 = \frac{398}{6} = 66.33, \bar{x}_2 = \frac{678}{10} = 67.80$$

- Sailors' mean = **66.33 in**
- Soldiers' mean = **67.80 in**

Step 3: Sample Variances (1 Mark)

$$s_1^2 = 28.67, s_2^2 = 17.07$$

Step 4: Pooled Variance (Equal Variances) (0.5 Mark)

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{5(28.67) + 9(17.07)}{14} = 21.21$$
$$s_p = \sqrt{21.21} = 4.60$$

Step 5: Test Statistic (0.5 Mark)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$
$$= \frac{66.33 - 67.80}{4.60\sqrt{\frac{1}{6} + \frac{1}{10}}} = \frac{-1.47}{2.37} = -0.62$$

Step 6: Degrees of Freedom & Critical Value (0.5 Mark)

$$df = n_1 + n_2 - 2 = 14$$

At α = 0.05 (one-tailed), $t_{critical} = -1.761$.

Step 7: Decision (0.5 Mark)

- $t_{Computed}$ = -0.62

- Since **-0.62 > -1.761**, we **fail to reject H_0**.

Q2. Given the dataset: [5, 7, 8, 5, 6, 7, 120, 6, 8, 7]. Apply the Interquartile Range (IQR) method to identify any outliers. Show the steps involved in calculating Q1, Q3, IQR, and the outlier boundaries, to identify the outliers. (3)

Answer:

Step 1: Sort the data **(0.5 Mark)**

[5, 5, 6, 6, 7, 7, 7, 8, 8, 120]

Step 2: Find Q1 and Q3 **(0.5 Mark)**

Q1 = 6 Q3 = 8

Step 3: Compute IQR and Determine bounds **(1.5 Mark)**

IQR = Q3 - Q1 = 8 - 6 = 2

Lower bound = Q1 - 1.5*IQR = 6 - 3 = 3

Upper bound = Q3 + 1.5*IQR = 8 + 3 = 11

Step 4: Identify outliers **(0.5 Mark)**

Any value < 3 or > 11 is an outlier. → Outlier = 120

Q3. Given the dataset:[5, 6, 7, 8, 5, 6, 7, 8, 50, 60]. Apply the skewness formula

$$Skewness = \left(\frac{\sqrt{n(n-1)}}{n-2}\right) \times \frac{\frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^3}{\left(\frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2\right)^{3/2}}$$

to calculate the skewness score of the dataset. Show all the steps and interpret whether the data is symmetrical, positively skewed, or negatively skewed. (3)

Answer:

Step 1:

$$n = 10$$

$$\bar{x} = 16.2$$

$$\frac{1}{n}\sum (x_i - \bar{x})^2 = \frac{3823.6}{10} = 382.36$$ **(0.5 Mark)**

$$\frac{1}{n}\sum (x_i - \bar{x})^3 = \frac{115049.76}{10} = 11504.976$$ **(0.5 Mark)**

$$\frac{\sqrt{n(n-1)}}{n-2} = \frac{\sqrt{10 \times 9}}{8} = \frac{\sqrt{90}}{8}$$

$$\frac{9.4868}{8} \approx 1.186$$ **0.5 Marks**

Ratio → 0.5 Mark

$$\frac{11504.976}{7476.6780383} \approx 1.5387818.$$

Final Skewness

1.824 **(0.5 Mark)**
The skewness score is +0.55, hence positively skewed **(0.5 Mark)**

Q4. Explain the methods to transform the data so that it closely approximates a normal distribution. (3)

Answer: Explain Log, exponential, Box-Cox techniques – Page 51,52 from reference book 1**(1 mark each)**

Q5. Explain the different methods of filling the missing values for the numerical data. (3)

Answer: Explain the following as given in Page 88 from reference book 2**(0.5 mark each)**

i. Ignore the tuple

ii. Fill manually

iii. Use global constant

iv. Central tendency

v. Attribute mean, median of same class

vi. Most probable value

Q6. The table shown has the data collected during the outbreak of smallpox.

	Attacked	Not Attacked
Vaccinated	31	469
Not Vaccinated	185	1315

Test the effectiveness of vaccination in preventing smallpox at 5% significance level . (3)

Answer: Attached as a separate document.

	Attacked (B)	Not attacked (b)	Total
vaccinated (A)	AB = 54	Ab = 446	500
Not vaccinated (a)	aB = 162	ab = 1338	1500
Total	216	1784	2000

Compute calculated χ^2

Group	Observed frequency O_{ij}	Expected frequency E_{ij}	$O_{ij} - E_{ij}$	$(O_{ij} - E_{ij})^2$	$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$
AB	31	54	-23	529	$\frac{529}{54} = 9.796$
Ab	469	446	23	529	$\frac{529}{446} = 1.186$
aB	185	162	23	529	$\frac{529}{162} = 3.265$
ab	1315	1338	-23	529	$\frac{529}{1338} = 0.395$

14.642

$$\chi^2 = 14.642 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Tabulated value is

DOF = $(r-1)(c-1) = (2-1)(2-1) = 1$ significance = 0.05

$\chi^2 = 3.841$ see from table.

Calculated χ^2 > Tabulated χ^2

$14.642 > 3.841$

Hence reject null hypothesis which states that vaccination & attack are dependente on each other & vaccination has been effective in preventing smallpox

Q7. Using the dataset of student *heights* (in cm) = {160, 165, 170, 175, 180} and *weights* (in kg) = {55, 58, 62, 65, 70}, calculate the covariance and interpret whether the relationship indicates a positive, negative, or no relation between the variables. (3)

Answer:

Step 1: Compute Means (0.5 marks)

$$\bar{A} = \frac{160 + 165 + 170 + 175 + 180}{5} = 170$$
$$\bar{B} = \frac{55 + 58 + 62 + 65 + 70}{5} = 62$$

Step 2: Compute E(A·B) =52885/5 =10577 (1 marks)

Step 3: Apply Formula (1 marks)

$$\mathrm{Cov}(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$
$$= 10577 - (170 \times 62)$$
$$= 10577 - 10540 = 37$$

There is a positive association between the variables. **(0.5 marks)**

Q8. A faculty member wants to analyse the performance of the students in the course he offers. The data set taken for analysis contains the following variables: Sl. No., Registration Number, Test1 marks, Test2 marks, Test3 marks, Final marks, and CGPA. Identify and explain the most suitable Dimensionality Reduction technique when the interpretability of the transformed result is not a concern. Further, identify the irrelevant features in this dataset. (2)

Answer: Identify PCA **(0.5 marks),** Explain PCA **(1 mark),** identify the irrelevant features **(0.5 marks)**

The most suitable dimensionality reduction technique in this context is **Principal Component Analysis (PCA).**

PCA transforms correlated performance measures (Test1, Test2, Test3, Final marks, CGPA) into a smaller set of uncorrelated components. The principal components are linear combinations of the original variables, and while they may be difficult to interpret directly, they capture most of the variance in the data. Since interpretability is not required here, the loss of direct meaning of components is acceptable. This reduces dimensionality, removes redundancy among features, and highlights underlying patterns in student performance.

Irrelevant Features: Sl. No., Registration Number

Q9. A researcher collects the following information from 20 individuals: Gender, Age, Height (in cm), Weight (in kg), Blood Group, Mobile Number, Education Level (High School, Undergraduate, Postgraduate, PhD), Monthly Income. Classify each variable into: (i) Discrete or Continuous (ii) Nominal, Ordinal, Interval, or Ratio. (2)

Answer: 0.25 mark each

Variable	**(i) Discrete / Continuous (ii) Scale (Nominal / Ordinal / Interval / Ratio)**

Variable	(i) Discrete / Continuous	(ii) Scale (Nominal / Ordinal / Interval / Ratio)
Gender	Discrete	Nominal
Age	Continuous	Ratio
Height	Continuous	Ratio
Weight	Continuous	Ratio
Blood Group	Discrete	Nominal
Mobile Number	Discrete	Nominal (identifier)
Education Level	Discrete	Ordinal
Monthly Income	Continuous	Ratio

Q10. A university is preparing student data for performance prediction. You observe the following problems:

1. The attribute attendance_percentage has missing values for several students.

2. The attribute grades are represented using both letter grades (A, B, C) and numerical scores (85, 72, 60). Explain the preprocessing steps you would apply to handle these issues. (2)

Answer:

1. Handling Missing Values in attendance_percentage

Mean/Median Imputation – Replace missing values with the mean or median of available attendance values if data is normally distributed or skewed.

Removal: If the number of missing cases is very small compared to dataset size, those rows can be dropped.

2. Handling Inconsistent Representation in grades

Standardization of Representation:

Option 1: Convert letters to numeric scores (e.g., A = 90–100, B = 75–89, C = 60–74, etc.).

Option 2: Convert numeric scores to letter grades (apply a grading scale to map numbers into categories).

Q11. Explain the Entity Identification problem. In a student dataset, both Roll Number and Registration Number are used as identifiers. Explain which attribute should be selected as the primary entity identifier during preprocessing and why. (2)

Answer: Page 94 in reference book 2

Entity Identification problem **(1 Mark)**, Identify and Justify primary entity **(1Mark)**